

SOFTWARE TESTING & VALIDATION · FINAL PROJECT

EasyService: Test Design Document — Part B (v1.0)

Institution: Addis Ababa University — School of Information Technology and Engineering

Course: Software Testing and Validation

Document: Test Design Document (Part B)

Repository: <https://github.com/Abyman2/Easyservice>

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1. Introduction & Traceability Matrix

This document details the formal test design derivations for EasyService based on Business Rules **BR-01 through BR-20**.

2. Equivalence Partitioning (EP)

2.1 Customer Identity & Registration (BR-01, BR-02)

- **Partition 1 (Valid Ethiopian):** Ethiopian Customer + Valid Fayda National ID $\rightarrow$ **VERIFIED** (Valid)
- **Partition 2 (Invalid Ethiopian):** Ethiopian Customer + Passport only $\rightarrow$ **REJECTED** (Invalid)
- **Partition 3 (Valid Foreigner):** Foreign Customer + Valid Passport $\rightarrow$ **VERIFIED** (Valid)
- **Partition 4 (Invalid Foreigner):** Foreign Customer + Fayda only $\rightarrow$ **REJECTED** (Invalid)

2.2 Payment Balance (BR-08, BR-09, BR-10)

- **Partition 1 (Insufficient):** $\text{Balance} < \text{Amount} \rightarrow$ Payment Declined (**EXPIRED** transaction)
- **Partition 2 (Exact Balance):** $\text{Balance} = \text{Amount} \rightarrow$ Payment Succeeded (**CONFIRMED** transaction)

- **Partition 3 (Sufficient Funds):** $\text{Balance} > \text{Amount}$ $\rightarrow$ Payment Succeeded (CONFIRMED transaction)

3. Boundary Value Analysis (BVA)

3.1 Listing Capacity & Booking Quantity (BR-06, BR-07)

For a listing capacity $C = 100$:

- $Q = 0$: Invalid (Boundary below minimum quantity) $\rightarrow$ Rejected
- $Q = 1$: Valid (Minimum quantity boundary) $\rightarrow$ Accepted
- $Q = 99$: Valid (Nominal interior value) $\rightarrow$ Accepted
- $Q = 100$: Valid (Exact remaining capacity boundary) $\rightarrow$ Accepted
- $Q = 101$: Invalid (Boundary above maximum capacity) $\rightarrow$ Rejected

3.2 Promotion Minimum Threshold (BR-11)

For a promotion with minimum purchase requirement $M = 500.00$:

- $A = 499.99$: Invalid (Boundary below threshold) $\rightarrow$ Discount Rejected
 - $A = 500.00$: Valid (Exact threshold boundary) $\rightarrow$ Discount Applied
 - $A = 500.01$: Valid (Boundary above threshold) $\rightarrow$ Discount Applied
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4. Decision Tables

4.1 Booking Eligibility Decision Table

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
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Customer Identity Verified?	**Yes**	**No**	Yes	Yes	Yes	Yes
Listing Published?	**Yes**	—	**No**	Yes	Yes	Yes
Quantity $\leq$ Available Stock?	**Yes**	—	—	**No**	Yes	Yes
Promo Code Valid?	**Yes**	—	—	—	**No**	Yes
Customer Balance $\geq$ Amount?	**Yes**	—	—	—	—	**No**
Actions						

Accept Booking (CONFIRMED)	\$\checkmark\$					
Reject Booking (Error/EXPIRED)		\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$	\$\checkmark\$

5. State Transition Testing (Transaction State Machine)



5.1 Valid State Transitions

- \$T_1\$: \$\text{PENDING} \rightarrow \text{Payment Success} \rightarrow \text{CONFIRMED}\$
- \$T_2\$: \$\text{CONFIRMED} \rightarrow \text{Service Complete} \rightarrow \text{COMPLETED}\$
- \$T_3\$: \$\text{PENDING} \rightarrow \text{Cancel} \rightarrow \text{CANCELLED}\$
- \$T_4\$: \$\text{CONFIRMED} \rightarrow \text{Cancel} \rightarrow \text{CANCELLED}\$ (Restores Inventory & Balance)
- \$T_5\$: \$\text{PENDING} \rightarrow \text{Payment Failure} \rightarrow \text{EXPIRED}\$

5.2 Explicit Negative State Transition Tests

- \$N_1\$: \$\text{COMPLETED} \rightarrow \text{Cancel} \rightarrow \text{CANCELLED} \rightarrow \text{FORBIDDEN}\$ (IllegalStateException)
- \$N_2\$: \$\text{COMPLETED} \rightarrow \text{Complete} \rightarrow \text{COMPLETED} \rightarrow \text{FORBIDDEN}\$ (IllegalStateException)
- \$N_3\$: \$\text{EXPIRED} \rightarrow \text{Confirm} \rightarrow \text{CONFIRMED} \rightarrow \text{FORBIDDEN}\$ (IllegalStateException)
- \$N_4\$: \$\text{CANCELLED} \rightarrow \text{Confirm} \rightarrow \text{CONFIRMED} \rightarrow \text{FORBIDDEN}\$ (IllegalStateException)

6. Derived Automated Test Cases

1. `createBooking_Valid_Success()`: Verifies \$T_1\$ valid transition.
2. `createBooking_UnverifiedUser_ThrowsException()`: Verifies Decision Table Rule 2.
3. `createBooking_QuantityExceedsCapacity_ThrowsException()`: Verifies BVA boundary \$Q = 101\$.
4. `calculateDiscountedTotal_BVA_Threshold()`: Verifies BVA values (\$499.99\$, \$500.00\$, \$500.01\$).
5. `cancelTransaction_AlreadyCompleted_ThrowsException()`: Verifies Negative State Transition \$N_1\$.